

UDAYAKUMAR P

| AI Researcher | Neuroimaging | Healthcare Data Science and Machine Learning | Expertise in Multi-model Data | Teaching | Trainer | Asst. Professor |

Objective

Experienced academic professional with over 10 years in Computer Science and Engineering, complemented by advanced expertise in AI, Machine Learning, and Deep Learning developed through PhD research—dedicated to fostering innovation and excellence in education and research.

Education

- 2021–2025 **PhD, Computer Science & Engineering**, *Vellore Institute of Technology*, Vellore.
Thesis Title: Human brain connectome mapping using Deep Learning (Thesis Submitted)
- 2008–2010: **Master of Engineering, Computer Science & Engineering**, *Anna University*, Tiruchirappalli.
CGPA : 8.05
- 2002–2006: **Bachelor of Engineering, Computer Science & Engineering**, *Anna University*, Chennai.
Percentage : 67 %

Teaching & Research Experience

Vellore Institute of Technology, Vellore, India

- December, 2021 **Teaching and Research Assistant (TRA)**.
– Present **Thesis Title:** Human brain Connectome mapping using Deep learning
Advisor : **Dr.R.Subhashini**, Associate Professor, School of Computer Science Engineering and Information Systems, Vellore Institute of Technology, Vellore, India

Wachemo University, Hosanna, Ethiopia

- November, 2016 **Lecturer, Department of Computer Science, College of Engineering and technology**.
– Sep, 2020

SRM-TRP Engineering College (SRM Group), Trichy, India

- June, 2013 – **Assistant Professor, Department of Computer Science & Engineering**.
May, 2016

Anna University, Chennai

- July, 2011 – **Teaching fellow, Department of Computer Technology, MIT campus**.
April, 2013

E.S College of Engineering & Technology, Tamilnadu, India

- June, 2010 – **Lecturer, Department of Computer Science & Engineering**.
June, 2011

Publications

Journal Article(Published)

- 2025 **Udayakumar P, R.Subhashini**, Connectome Graph Attention Network (CGAT): A Multimodal approach for enhanced classification of Alzheimer's disease controls, In **Results in Engineering,2025 (Impact Fator: 7.9)**.
- 2025 **Udayakumar P, R.Subhashini**, Integrating deep learning in brain connectome mapping: Insights from a systematic review, In **International Research Journal of Multidisciplinary Scope (IRJMS),2025; 6(2): 1-20**.
- 2024 **Udayakumar P, R.Subhashini**, Connectome based schizophrenia prediction using structural connectivity - Deep Graph Neural Network(sc-DGNN), In **Journal of X-Ray Science and Technology (Impact Factor: 3.0)**.
- 2024 **Udayakumar P, M.Indhumathi**, Semantic web based sensor planning services (SPS) for sensor web enablement (SWE), In **International Journal of Computer Engineering Science (IJCES) <https://doi.org/10.48550/arXiv.1207.5310>, 2012** .

Communicated Journal Article

- 2025 **Udayakumar P, R.Subhashini**, CST-GCNN: Spatio-Temporal Connectome Graph Network for Schizophrenia Classification, In *IEEE Access* .

In Conference Proceedings

- 2024 P Udayakumar and R. Subhashini. fmri data to connectivity matrices: Building functional connectomes for deep learning models. In *IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering 2024 (IEEE WIECON-ECE 2024)*, pages 1–5. IEEE, 2024.
- 2023 P Udayakumar and R. Subhashini. Diffusion mri preprocessing for deep learning analysis of brain structural connectivity. In *2023 12th International Conference on Advanced Computing (ICoAC)*, pages 1–5. IEEE, 2023.

Fellowships & Awards

- 2025 **Raman Research Award** of Vellore Institute of Technology, Vellore,India, as a PhD research scholar.
- 2024 **Raman Research Award** of Vellore Institute of Technology, Vellore,India, as a PhD research scholar.
- 2024 **Dr.APJ Abdul Kalam Award** of Vellore Institute of Technology, Vellore, India, as a PhD research scholar.

Technical skills

Programming Languages	Python, C, C++, Advanced JAVA
Web Technologies	HTML 5, PHP, JSP, , ReactJS
Database	SQL, MySQL
Brain Imaging Tools	Matlab, FSL, FreeSurfer, SPM12, MRtrix, ANTs, DSISudio
ML Frameworks	PyTorch, Keras, TensorFlow, PyTorch Lightning, PyG
Data Visualization	NumPy, Pandas, Matplotlib

AI Specializations LLM, SSL, multi-modal AI models, neural networks, generative models
MLOps Hugging Face, DeepSpeed, Megatron

Certification

Jan, 2025 : **Introduction to Machine Learning by NPTEL**, IIT Ropar.
Jul, 2024 : **Oracle Cloud Infrastructure Generative AI Certified Professional**, Oracle.
Jan, 2023 : **Deep Learning by NPTEL**, IIT Ropar.

Key Achievements

- Developed scalable AI algorithms for brain imaging and healthcare applications.
- Published articles on deep learning models applied to neuroimaging data.
- Led student research projects in multi-modal AI, resulting in strong student engagement.

Research Interests

Large-Scale Healthcare Data Analysis
Generative AI Models in Clinical Applications
Integration of Deep Learning in Multi-modal Medical Imaging

Referees

Dr.R.S.Somasundaram
Ph.D.,M.E.,MCA.,M.Phil.,
Professor & Head, Department of
Computer Science & Engineering
Coimbatore Institute of Technology,India
☎ +(91) 9486978727
✉ somasundaram.rs@cit.edu.in

Dr. S.Neelavathy Pari B.E., M.E., Ph.D.,
Associate Professor, Department of
Computer Science & Engineering
Anna university, Chennai, India
☎ +(91) 9962827638
✉ neela@annauniv.edu

Dr. R.Subhashini B.E., M.E., Ph.D.,
Associate Professor, School of
Computer Science Engineering & Information
Systems(SCORE)
Vellore Institute of Technology, Vellore,India
☎ +(91) 9003991299
✉ rsubhashini@vit.ac.in